

Table F1. **The landscape: the 323 works outside the training core**, grouped by the stage at which calibration enters (Table A1) and then by method family, oldest first; the 80 core works and the 5 theory and analysis works are listed in Table C1. Yr: year of the first public version (a citation may show a later venue year); Contract: the form of the output the confidence is attached to: typed (a probability per declared option), set (a prediction set), constrained (one output under a grammar or schema), score (one confidence number per answer, stated, read off or estimated by a separate model) or free-text (free text or a label with no per-answer probability; any confidence is a verbal hedge, an abstention, or agreement across sampled answers); Access: white (needs logits or weights), black (text only) or either; Samples: the method needs repeated inference; Guar.: a stated distribution-free guarantee; Trained: yes when calibration is a training objective of the model itself, partial when a separate model (a probe, calibrator, router or reward model) is trained; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric); Task: the task evaluated; Action: what an explicit rule or a trained behaviour does with the confidence when it is low (abstain; hand off = pass the case to another model or a human, which the papers call routing, cascading, deferral or escalation; retrieve = trigger retrieval; filter = remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers; adapt compute = keep sampling, reasoning, querying or exploring, or spend a larger compute budget (the same rule stops early when the confidence is high); set = return a prediction set; none = no rule or trained behaviour acts on the confidence, including a selective-prediction curve reported only as an evaluation). ? = not determinable from the paper.

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
S0 read-off (25)									
<i>Calibration after alignment</i>									
OpenAI [2023]	2023	typed	white	no	no	no	ECE	QA	none
Zhu et al. [2023]	2023	score	white	no	no	no	ECE	QA	none
Huang et al. [2026]	2026	typed	white	no	no	no	ECE	QA	none
Proskurina et al. [2026]	2026	typed	white	no	no	no	ECE	QA	none
Sanz-Guerrero et al. [2026]	2026	typed	white	no	no	no	multiple	QA	none
<i>Empirical studies of read-off calibration</i>									
Desai and Durrett [2020]	2020	typed	white	no	no	no	ECE	classification	none
Wang et al. [2020]	2020	score	white	no	no	no	ECE	generation	none
Dan and Roth [2021]	2021	typed	white	no	no	no	ECE	classification	none
Xiao and Wang [2021]	2021	score	white	no	no	no	other	generation	none
Ahuja et al. [2022]	2022	typed	white	no	no	no	ECE	classification	none
Chen et al. [2023a]	2022	typed	white	no	no	no	ECE	classification	none
Si et al. [2022]	2022	score	white	no	no	no	ECE	QA	none
Stengel-Eskin and Van Durme [2023]	2022	constrained	white	no	no	no	ECE	code	none
Xiao et al. [2022]	2022	typed	white	no	no	no	multiple	classification	none
Li et al. [2025h]	2023	typed	white	no	no	no	multiple	classification	none
Zhang et al. [2023d]	2023	typed	white	no	no	no	ECE	classification	none
Mahaut et al. [2024]	2024	typed	either	no	no	no	other	QA	none
Stolfo et al. [2024]	2024	score	white	no	no	no	other	generation	none
Joshi et al. [2025]	2025	typed	white	no	no	no	ECE	QA	none
<i>Likelihood and logit read-off</i>									
Ott et al. [2018]	2018	score	white	yes	no	no	other	generation	none
Lovering et al. [2025]	2024	typed	white	no	no	no	other	reasoning	none
Plaut et al. [2024]	2024	typed	white	no	no	no	multiple	QA	abstain
Spieß et al. [2024]	2024	score	white	no	no	no	multiple	code	none
Wang et al. [2024d]	2024	typed	white	no	no	no	none	QA	none
<i>Self-evaluation and read-off calibration</i>									
Kadavath et al. [2022]	2022	score	white	no	no	partial	multiple	QA	none
S1 post-hoc map (52)									
<i>Fixed maps over the read-off</i>									
Zhang et al. [2023c]	2023	score	white	no	no	no	other	generation	none
Bakman et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Fadeeva et al. [2024]	2024	score	white	no	no	no	AUROC	generation	none
Li et al. [2025b]	2025	score	white	no	no	no	multiple	reasoning	none
Vazhentsev et al. [2026]	2025	score	white	no	no	no	AUROC	generation	none
Badhe et al. [2026]	2026	typed	white	no	no	no	multiple	classification	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
<i>Label-bias and option-prior debiasing</i>									
Zheng et al. [2023]	2023	typed	white	no	no	no	none	QA	none
Reif and Schwartz [2024]	2024	typed	white	no	no	no	none	classification	none
<i>Learned calibrators</i>									
Ye and Durrett [2022]	2021	score	black	no	no	partial	multiple	QA	none
Zhang et al. [2021]	2021	score	white	no	no	partial	selective	QA	filter
Dhuliawala et al. [2022]	2022	score	white	no	no	partial	multiple	QA	none
Han et al. [2022]	2022	typed	white	no	no	partial	none	classification	none
Liu et al. [2024a]	2024	score	white	no	no	partial	multiple	QA	none
Liu et al. [2024d]	2024	score	either	no	no	partial	AUROC	QA	none
Ulmer et al. [2024a]	2024	score	black	no	no	partial	multiple	QA	none
Radharapu et al. [2025]	2025	score	white	no	no	partial	ECE	judging	none
Luo et al. [2026]	2026	typed	white	no	no	no	multiple	QA	none
<i>Multicalibration and task-specific recalibration</i>									
Detommaso et al. [2024]	2024	score	either	no	no	no	multiple	QA	none
Liu et al. [2024b]	2024	typed	white	no	no	no	ECE	classification	none
<i>Probes and learned confidence heads</i>									
Kumar and Sarawagi [2019]	2019	score	white	no	no	partial	ECE	generation	none
Mielke et al. [2022]	2020	free-text	either	no	no	partial	multiple	QA	none
Burns et al. [2022]	2022	typed	white	no	no	partial	other	classification	none
Azaria and Mitchell [2023]	2023	typed	white	no	no	partial	AUROC	classification	none
CH-Wang et al. [2024]	2023	score	white	no	no	partial	other	generation	none
Liu et al. [2023a]	2023	typed	white	no	no	partial	ECE	classification	none
Liu et al. [2023c]	2023	score	white	no	no	partial	multiple	generation	none
Slobodkin et al. [2023]	2023	free-text	white	no	no	partial	other	QA	none
Beigi et al. [2024a]	2024	score	white	no	no	partial	ECE	QA	none
Du et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Gottesman and Geva [2024]	2024	score	white	no	no	partial	other	QA	none
Ji et al. [2024]	2024	score	white	no	no	partial	other	generation	none
Kossen et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Orgad et al. [2024]	2024	score	white	no	no	partial	AUROC	QA	none
Su et al. [2024b]	2024	score	white	no	no	partial	AUROC	generation	none
Khanmohammadi et al. [2025]	2025	score	white	no	no	partial	multiple	QA	none
Lu [2026]	2026	score	white	no	no	partial	AUROC	reasoning	none
<i>Prompt and context calibration</i>									
Holtzman et al. [2021]	2021	typed	white	no	no	no	none	classification	none
Zhao et al. [2021]	2021	typed	white	no	no	no	none	classification	none
Fei et al. [2023]	2023	typed	white	no	no	no	none	classification	none
Jiang et al. [2023b]	2023	typed	white	yes	no	no	none	classification	none
Liusie et al. [2023]	2023	typed	white	no	no	no	none	classification	none
Zhou et al. [2023b]	2023	typed	white	no	no	no	none	classification	none
Abbas et al. [2024]	2024	typed	white	no	no	no	ECE	classification	none
Cho et al. [2025]	2024	typed	white	no	no	no	none	classification	none
Gundem et al. [2025]	2025	typed	white	no	no	no	none	classification	none
<i>Scaling and binning</i>									
Guo et al. [2017]	2017	typed	white	no	no	no	multiple	classification	none
Kull et al. [2019]	2019	typed	white	no	no	no	multiple	classification	none
Kumar et al. [2019]	2019	typed	either	no	yes	no	multiple	classification	none
Gupta and Ramdas [2021]	2021	typed	either	no	yes	no	ECE	classification	none
Shen et al. [2024]	2024	typed	white	no	no	partial	multiple	QA	none
Xie et al. [2024a]	2024	score	white	no	no	partial	multiple	QA	none
Luo et al. [2025]	2025	typed	white	no	no	no	multiple	QA	none
S2 prompt elicitation (54)									
<i>Abstention prompting</i>									
Zong et al. [2026]	2026	score	black	no	no	no	multiple	QA	abstain
<i>Introspection, deliberation and task-specific elicitation</i>									
Sun et al. [2024]	2024	constrained	either	no	no	partial	multiple	other	none
Yang et al. [2024c]	2024	score	black	yes	no	no	multiple	QA	none
Mei et al. [2025]	2025	score	black	no	no	no	ECE	reasoning	none
<i>Linguistic hedges and epistemic markers</i>									
Sileo and Moens [2022]	2022	free-text	white	no	no	no	none	reasoning	none
Ren et al. [2023b]	2023	score	white	yes	no	no	multiple	QA	none
Singh et al. [2023]	2023	score	black	no	no	no	other	other	none
Yin et al. [2023]	2023	free-text	black	no	no	no	other	QA	none
Ghafari et al. [2024]	2024	free-text	either	no	no	partial	other	QA	none
Huang et al. [2024b]	2024	score	black	yes	no	no	multiple	generation	hand off
Kumar et al. [2024]	2024	score	white	no	no	no	other	QA	none
Li et al. [2024a]	2024	free-text	black	no	no	no	other	reasoning	filter
Li et al. [2024c]	2024	score	either	?	no	no	multiple	QA	none
Pawitan and Holmes [2024]	2024	score	black	no	no	no	ECE	reasoning	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Sicilia et al. [2024]	2024	score	either	no	no	partial	Brier	QA	none
Tang et al. [2024]	2024	free-text	black	no	no	no	other	other	none
Xue et al. [2024]	2024	score	either	?	no	no	multiple	QA	none
Zhao et al. [2024b]	2024	score	black	no	no	no	ECE	QA	retrieve
Liu et al. [2025e]	2025	free-text	black	no	no	no	ECE	QA	none
Sun et al. [2025]	2025	score	black	no	no	no	ECE	reasoning	none
Liu and Cohan [2026]	2026	free-text	black	no	no	no	other	QA	none
<i>Self-evaluation prompts</i>									
Tian et al. [2025]	2025	score	black	no	no	no	multiple	judging	none
<i>Stated confidence that triggers an action</i>									
Ni et al. [2024a]	2024	free-text	black	no	no	no	other	QA	retrieve
Ferrer et al. [2026]	2026	score	black	no	no	no	multiple	judging	hand off
<i>Verbalised numeric confidence</i>									
Tanneru et al. [2023]	2023	score	black	yes	no	no	other	QA	none
Tian et al. [2023]	2023	score	black	no	no	no	multiple	QA	none
Xiong et al. [2023]	2023	score	black	yes	no	no	multiple	reasoning	none
Zhou et al. [2023a]	2023	free-text	black	no	no	no	ECE	QA	none
Belem et al. [2024]	2024	free-text	black	no	no	no	other	other	none
Groot and Valdenegro-Toro [2024]	2024	score	black	no	no	no	ECE	other	none
Lee et al. [2024a]	2024	free-text	black	no	no	no	none	judging	none
Ni et al. [2024b]	2024	score	either	no	no	no	other	QA	none
Rivera et al. [2024]	2024	score	black	yes	no	no	multiple	classification	none
Wang et al. [2024f]	2024	score	black	no	no	no	multiple	classification	none
Wei et al. [2024]	2024	score	black	yes	no	no	ECE	QA	none
Yang et al. [2024d]	2024	score	black	no	no	no	ECE	QA	none
Yona et al. [2024]	2024	free-text	black	yes	no	no	other	QA	none
Chhikara [2025]	2025	score	black	no	no	no	ECE	QA	none
Liu et al. [2025d]	2025	free-text	black	yes	no	no	other	QA	none
Naderi et al. [2025]	2025	score	black	no	no	no	multiple	QA	none
Obadinma and Zhu [2025]	2025	score	black	no	no	no	other	QA	none
Tao et al. [2025b]	2025	free-text	black	no	no	partial	multiple	QA	none
Tao et al. [2025a]	2025	score	either	no	no	no	multiple	QA	none
Wang and Stengel-Eskin [2025]	2025	score	black	yes	no	no	multiple	QA	none
Wang et al. [2025d]	2025	score	black	no	no	no	multiple	QA	none
Yoon et al. [2025]	2025	score	black	no	no	no	multiple	reasoning	none
Yu et al. [2025b]	2025	score	black	yes	no	no	multiple	agentic	none
Zhou et al. [2025]	2025	score	black	yes	no	no	multiple	QA	none
Cacioli [2026]	2026	score	black	no	no	no	AUROC	QA	none
French Constant et al. [2026]	2026	score	black	no	no	no	Brier	QA	none
Hsiao [2026]	2026	score	black	no	no	no	ECE	judging	none
Kumaran et al. [2026]	2026	score	white	no	no	no	multiple	QA	none
Marashian et al. [2026]	2026	score	black	no	no	no	ECE	generation	none
Yang et al. [2026a]	2026	score	black	no	no	no	multiple	QA	none
S3 extra inference (121)									
<i>Benchmarks and empirical studies</i>									
Fadeeva et al. [2023]	2023	free-text	either	?	no	no	AUROC	generation	none
Huang et al. [2025c]	2023	free-text	either	yes	no	no	AUROC	other	none
Huang et al. [2024a]	2024	free-text	either	yes	no	no	AUROC	QA	none
Ling et al. [2024]	2024	typed	white	yes	no	no	AUROC	classification	none
Vashurin et al. [2025]	2024	free-text	either	?	no	no	multiple	generation	none
<i>Conformal abstention, selection and cascades</i>									
Fisch et al. [2020]	2020	set	white	no	yes	no	coverage	QA	set
Lee et al. [2024b]	2023	free-text	either	no	yes	partial	coverage	generation	abstain
Yadkori et al. [2024b]	2024	free-text	black	yes	yes	no	coverage	QA	abstain
Gui et al. [2024]	2024	free-text	either	yes	yes	partial	coverage	generation	abstain
Jung et al. [2024]	2024	typed	either	yes	yes	no	multiple	judging	hand off
Tayebati et al. [2025]	2025	set	white	no	no	partial	multiple	QA	abstain
<i>Conformal factuality and claim filtering</i>									
Cherian et al. [2024]	2024	free-text	either	yes	yes	partial	coverage	generation	filter
Mohri and Hashimoto [2024]	2024	free-text	either	yes	yes	no	coverage	generation	filter
Feng et al. [2025]	2025	free-text	black	yes	yes	no	coverage	QA	filter
Jiang et al. [2025]	2025	free-text	black	yes	yes	partial	coverage	QA	filter
Lee et al. [2025]	2025	free-text	either	?	yes	no	coverage	QA	abstain
Rubin-Toles et al. [2025]	2025	free-text	black	yes	yes	no	coverage	reasoning	filter
<i>Conformal prediction sets</i>									
Dey et al. [2022]	2021	set	white	no	yes	no	coverage	classification	set
Fisch et al. [2021]	2021	set	white	no	yes	partial	coverage	classification	set
Choubey et al. [2022]	2022	set	white	no	yes	no	coverage	classification	set
Deutschmann et al. [2023]	2023	set	white	no	yes	no	coverage	generation	set
Kumar et al. [2023]	2023	set	white	no	yes	no	coverage	QA	set

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Li et al. [2024d]	2023	set	either	yes	yes	no	coverage	QA	set
Quach et al. [2023]	2023	set	either	yes	yes	no	coverage	generation	set
Ravfogel et al. [2023]	2023	set	white	no	yes	no	coverage	generation	set
Ren et al. [2023a]	2023	set	white	no	yes	no	coverage	robotics	hand off
Wang et al. [2025f]	2023	set	white	no	yes	no	coverage	robotics	hand off
Zerva and Martins [2024]	2023	set	white	?	yes	no	coverage	judging	set
Kaur et al. [2024]	2024	set	black	yes	yes	no	multiple	QA	set
Kostumov et al. [2024]	2024	set	white	no	yes	no	multiple	QA	set
Liang et al. [2024b]	2024	set	white	no	yes	no	coverage	robotics	hand off
Liu and Wu [2024]	2024	free-text	black	yes	yes	partial	multiple	generation	filter
Ren et al. [2024]	2024	set	white	no	yes	no	other	robotics	adapt compute
Su et al. [2024a]	2024	set	black	yes	yes	no	coverage	QA	set
Ulmer et al. [2024b]	2024	set	white	no	yes	no	coverage	generation	set
Vishwakarma et al. [2024]	2024	set	white	no	yes	partial	coverage	QA	set
Wang et al. [2024a]	2024	set	black	yes	yes	no	multiple	QA	set
Wang et al. [2025c]	2024	set	white	no	yes	no	coverage	robotics	hand off
Ye et al. [2024]	2024	set	white	no	yes	no	multiple	QA	set
Zhang et al. [2024b]	2024	set	either	no	yes	no	coverage	QA	set
Chan et al. [2025]	2025	set	?	?	yes	no	coverage	QA	adapt compute
Doula et al. [2025]	2025	set	black	yes	yes	no	coverage	robotics	hand off
Huang et al. [2025b]	2025	set	white	no	yes	no	multiple	QA	hand off
Kaur et al. [2025]	2025	set	black	yes	yes	no	none	reasoning	set
Kuwahara et al. [2025]	2025	free-text	either	no	yes	no	coverage	generation	filter
Noorani et al. [2025]	2025	set	black	yes	yes	no	coverage	generation	set
Shahrokhi et al. [2025]	2025	set	black	yes	yes	no	coverage	code	set
Sheng et al. [2025]	2025	set	either	no	yes	no	coverage	judging	set
Sundarsingh et al. [2025]	2025	set	?	?	yes	no	coverage	robotics	hand off
Wang et al. [2025g]	2025	set	black	yes	yes	no	coverage	QA	abstain
Yang and Liu [2025]	2025	set	black	yes	yes	no	multiple	QA	set
Emmenegger et al. [2026]	2026	free-text	?	yes	yes	no	coverage	generation	abstain
Zhang et al. [2026b]	2026	set	?	?	yes	no	none	robotics	adapt compute
<i>Ensembles and cross-model disagreement</i>									
Dong et al. [2018]	2018	constrained	white	yes	no	partial	selective	code	none
Hou et al. [2024]	2023	free-text	black	yes	no	no	AUROC	QA	hand off
Tonolini et al. [2024]	2024	typed	black	yes	no	no	multiple	classification	none
Kruse et al. [2025]	2025	typed	?	yes	no	no	multiple	classification	none
Hamidieh et al. [2026]	2026	free-text	black	yes	no	no	multiple	generation	none
<i>Internal-state signals</i>									
Fomicheva et al. [2020]	2020	score	white	yes	no	no	other	generation	none
Chen et al. [2024]	2024	free-text	white	yes	no	no	AUROC	QA	none
Li et al. [2026e]	2025	free-text	black	yes	no	no	AUROC	QA	none
<i>Long-form and claim-level</i>									
Da et al. [2024]	2024	free-text	black	yes	no	no	multiple	QA	none
Jiang et al. [2024]	2024	free-text	black	yes	no	no	AUROC	generation	filter
Zhang et al. [2025d]	2024	free-text	either	yes	no	no	multiple	generation	filter
Zhang et al. [2024c]	2024	free-text	black	yes	no	no	selective	generation	filter
Fan et al. [2026]	2026	free-text	black	yes	no	no	AUROC	generation	none
<i>Risk control and early exit</i>									
Angelopoulos et al. [2025]	2021	set	either	no	yes	no	coverage	other	set
Schuster et al. [2021]	2021	typed	white	no	yes	partial	coverage	classification	adapt compute
Angelopoulos et al. [2022]	2022	set	either	no	yes	no	coverage	other	set
Schuster et al. [2022]	2022	score	white	no	yes	partial	other	generation	adapt compute
Zollo et al. [2023]	2023	free-text	black	no	yes	no	coverage	generation	none
Jazbec et al. [2024]	2024	typed	white	no	yes	no	multiple	other	adapt compute
Kang et al. [2024]	2024	set	black	yes	yes	no	coverage	generation	set
Kladny et al. [2024]	2024	set	black	yes	yes	no	coverage	generation	set
Wang et al. [2024b]	2024	set	black	yes	yes	no	coverage	QA	set
Chen et al. [2025b]	2025	free-text	black	yes	yes	no	coverage	generation	filter
Ke et al. [2025]	2025	set	black	yes	yes	no	multiple	QA	set
Overman and Bayati [2025]	2025	score	black	no	yes	no	coverage	QA	hand off
Wang et al. [2025b]	2025	set	black	yes	yes	no	coverage	QA	abstain
Ye [2025]	2025	set	black	yes	yes	no	coverage	QA	set
Feng et al. [2026]	2026	constrained	either	?	yes	no	other	agentic	hand off
Wang et al. [2026b]	2026	score	white	no	yes	no	coverage	reasoning	abstain
<i>Sampling consistency</i>									
Wang et al. [2023b]	2022	free-text	black	yes	no	no	other	reasoning	none
Agrawal et al. [2023]	2023	free-text	black	yes	no	no	multiple	other	none
Chen and Mueller [2023]	2023	free-text	black	yes	no	no	AUROC	QA	filter
Cohen et al. [2023]	2023	free-text	black	yes	no	no	other	QA	none
Cole et al. [2023]	2023	free-text	black	yes	no	no	multiple	QA	abstain

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Friel and Sanyal [2023]	2023	free-text	black	yes	no	no	AUROC	judging	none
Liu et al. [2023b]	2023	typed	either	yes	no	no	other	judging	none
Manakul et al. [2023]	2023	free-text	black	yes	no	no	other	generation	none
Mündler et al. [2023]	2023	free-text	black	yes	no	no	other	generation	filter
Portillo Wightman et al. [2023]	2023	typed	either	yes	no	no	ECE	QA	none
Zhao et al. [2024a]	2023	free-text	black	yes	no	no	other	QA	none
Becker and Soatto [2024]	2024	typed	white	yes	no	no	multiple	QA	none
Gao et al. [2024]	2024	free-text	black	yes	no	no	ECE	QA	none
Lyu et al. [2025]	2024	free-text	black	yes	no	no	multiple	reasoning	none
Yadkori et al. [2024a]	2024	free-text	black	yes	no	no	selective	QA	abstain
Yang et al. [2024a]	2024	typed	black	yes	no	no	multiple	QA	none
Zhang et al. [2024a]	2024	typed	either	yes	no	no	ECE	QA	none
Tan et al. [2025]	2025	free-text	white	yes	no	partial	AUROC	QA	none
<i>Semantic entropy and meaning clustering</i>									
Kuhn et al. [2023]	2023	free-text	white	yes	no	no	AUROC	QA	none
Lin et al. [2024]	2023	free-text	black	yes	no	no	multiple	QA	none
Zhang et al. [2023b]	2023	free-text	black	yes	no	no	AUROC	QA	none
Aichberger et al. [2025]	2024	free-text	white	yes	no	no	AUROC	QA	none
Ao et al. [2024]	2024	free-text	black	yes	no	no	multiple	QA	none
Farquhar et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Grewal et al. [2024]	2024	free-text	either	yes	no	partial	AUROC	QA	none
Nikitin et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Qiu and Miikkulainen [2024]	2024	free-text	white	yes	no	no	AUROC	QA	none
Zhang et al. [2024e]	2024	free-text	black	yes	no	no	other	QA	none
Ciosek et al. [2025]	2025	free-text	?	yes	no	no	AUROC	QA	adapt compute
Ma et al. [2025]	2025	free-text	white	yes	no	no	AUROC	QA	none
Nguyen et al. [2025]	2025	free-text	black	yes	no	no	AUROC	generation	none
Sharma and David [2025]	2025	free-text	?	yes	no	no	other	code	abstain
Zhao et al. [2026]	2025	free-text	black	yes	no	no	multiple	QA	none
Bouchard et al. [2026]	2026	free-text	black	yes	no	no	AUROC	code	none
He et al. [2026a]	2026	free-text	black	yes	no	no	AUROC	code	abstain
Sun et al. [2026a]	2026	free-text	?	yes	no	no	AUROC	QA	adapt compute
<i>Token-level aggregation</i>									
Malinin and Gales [2020]	2020	free-text	white	yes	no	no	AUROC	generation	none
Duan et al. [2024]	2023	free-text	white	yes	no	no	AUROC	QA	none
Wang et al. [2025a]	2024	free-text	white	yes	no	no	AUROC	QA	filter
S4 training (80): listed in Table C1									
Theory and analysis (5): listed in Table C1									
X consumers and contract (71): output-contract works detailed in Table D1									
<i>Abstention</i>									
Kamath et al. [2020]	2020	score	white	no	no	partial	selective	QA	abstain
Feng et al. [2024]	2024	free-text	black	no	no	no	ECE	QA	abstain
Srinivasan et al. [2024]	2024	score	white	no	no	no	multiple	QA	adapt compute
<i>Adaptive compute</i>									
Aggarwal et al. [2023]	2023	free-text	black	yes	no	no	none	reasoning	adapt compute
Jiang et al. [2023a]	2023	score	white	no	no	no	none	QA	retrieve
Damani et al. [2024]	2024	score	white	yes	no	partial	multiple	reasoning	hand off
Li et al. [2024f]	2024	free-text	black	yes	no	no	none	reasoning	adapt compute
Fu et al. [2025]	2025	score	white	yes	no	no	other	reasoning	filter
Yang et al. [2025b]	2025	score	white	no	no	no	other	reasoning	adapt compute
<i>Confidence-weighted selection</i>									
Taubenfeld et al. [2025]	2025	score	either	yes	no	no	multiple	reasoning	filter
<i>Constrained decoding</i>									
Scholak et al. [2021]	2021	constrained	white	no	no	no	none	code	none
Beurer-Kellner et al. [2023]	2022	constrained	white	no	no	no	none	generation	none
Poesia et al. [2022]	2022	constrained	white	no	no	partial	none	code	none
Geng et al. [2023]	2023	constrained	white	no	no	no	none	other	none
Lew et al. [2023]	2023	constrained	white	yes	no	no	none	generation	none
Willard and Louf [2023]	2023	constrained	white	no	no	no	none	generation	none
Beurer-Kellner et al. [2024]	2024	constrained	white	no	no	no	none	generation	none
Dong et al. [2025]	2024	constrained	white	no	no	no	none	generation	none
Geng et al. [2024b]	2024	constrained	black	no	no	no	none	other	none
Koo et al. [2024]	2024	constrained	white	no	no	no	none	generation	none
Park et al. [2024]	2024	constrained	white	yes	no	no	none	code	none
Ugare et al. [2024]	2024	constrained	white	no	no	no	none	code	none
Banerjee et al. [2025]	2025	constrained	white	no	no	no	none	reasoning	none
Li et al. [2026b]	2025	constrained	white	no	no	no	none	code	none
Loula et al. [2025]	2025	constrained	white	yes	no	no	other	code	none

Table F1, continued

Work	Yr	Contract	Access	Samples	Guar.	Trained	Metric	Task	Action
Mündler et al. [2025]	2025	constrained	white	no	no	no	none	code	none
Park et al. [2025b]	2025	constrained	white	no	no	no	none	generation	none
Wang et al. [2025c]	2025	free-text	either	no	no	partial	none	generation	none
Reddy et al. [2026]	2026	constrained	white	no	no	no	none	reasoning	none
<i>Deferral</i>									
Jitkrittum et al. [2023]	2023	typed	white	no	no	partial	none	classification	hand off
Fanconi and van der Schaar [2025]	2025	score	either	?	no	partial	none	QA	hand off
Wu et al. [2025b]	2025	score	black	?	no	no	other	reasoning	hand off
Kondadadi and Ortega [2026]	2026	typed	either	?	no	partial	none	classification	hand off
<i>Human reliance on model output and stated confidence</i>									
Si et al. [2024]	2023	free-text	black	no	no	no	other	QA	none
Kim et al. [2024]	2024	free-text	black	no	no	no	other	QA	none
Steyvers et al. [2025b]	2024	free-text	either	no	no	no	multiple	QA	none
Zhou et al. [2024b]	2024	free-text	black	no	no	no	other	QA	none
Zhou et al. [2024a]	2024	free-text	black	no	no	no	other	QA	none
Rathi et al. [2025]	2025	free-text	black	no	no	no	other	QA	none
Villavicencio et al. [2026]	2026	score	either	?	no	no	other	QA	none
<i>Judge calibration</i>									
Liu et al. [2023d]	2023	score	black	no	no	no	none	judging	none
Wang et al. [2023a]	2023	free-text	black	yes	no	no	none	judging	hand off
Lee et al. [2026b]	2025	free-text	black	no	no	no	coverage	judging	none
Li et al. [2026c]	2026	typed	black	no	no	no	multiple	judging	hand off
<i>Routing/cascade</i>									
Varshney and Baral [2022]	2022	typed	white	no	no	no	none	classification	hand off
Aggarwal et al. [2024]	2023	score	black	yes	no	partial	none	QA	hand off
Chen et al. [2023b]	2023	score	black	no	no	partial	none	QA	hand off
Yue et al. [2024]	2023	free-text	black	yes	no	no	other	reasoning	hand off
Ding et al. [2024]	2024	score	black	no	no	partial	none	generation	hand off
Gupta et al. [2024]	2024	score	white	no	no	partial	none	generation	hand off
Hu et al. [2024]	2024	score	black	no	no	partial	none	QA	hand off
Narasimhan et al. [2024]	2024	score	white	no	no	no	none	generation	hand off
Ong et al. [2024]	2024	score	black	no	no	partial	none	generation	hand off
Ramirez et al. [2024]	2024	typed	white	no	no	no	none	classification	hand off
Soiffer et al. [2025]	2025	free-text	black	yes	no	no	none	generation	hand off
Zellinger and Thomson [2025]	2025	typed	white	no	no	no	ECE	QA	hand off
Bouchard [2026]	2026	score	either	?	no	no	other	QA	hand off
<i>Structured-output benchmark</i>									
Tang et al. [2023]	2023	constrained	either	no	no	no	none	generation	none
Shorten et al. [2024]	2024	constrained	black	no	no	no	none	QA	none
Tam et al. [2024]	2024	constrained	either	no	no	no	none	reasoning	none
Xia et al. [2024a]	2024	constrained	black	no	no	no	none	generation	none
Geng et al. [2025]	2025	constrained	white	no	no	no	none	generation	none
Yang et al. [2025a]	2025	constrained	black	no	no	no	none	generation	none
Fan [2026]	2026	constrained	black	no	no	no	none	reasoning	none
Galeone et al. [2026]	2026	constrained	black	no	no	no	none	reasoning	none
Lee et al. [2026a]	2026	constrained	either	no	no	no	none	reasoning	none
Singh et al. [2026a]	2026	constrained	black	no	no	no	none	QA	none
<i>Studies and benchmarks of abstention</i>									
Tomani et al. [2024]	2024	free-text	either	yes	no	no	multiple	QA	none
Kirichenko et al. [2025]	2025	free-text	black	no	no	no	other	reasoning	none
Wang et al. [2026c]	2026	score	black	no	no	no	multiple	QA	abstain
<i>Typed classification</i>									
Wang et al. [2024c]	2024	typed	white	no	no	no	none	classification	none